Differentiability is one of the most important ideas in calculus, but it can also be difficult to understand when it is only presented through formulas. A derivative represents the instantaneous rate of change of a function at a specific point, but that idea can feel abstract when it is only shown symbolically. The purpose of this project was to create an interactive application that makes differentiability easier to visualize.

I created an R Shiny application that allows a user to examine a mathematical function graphically and numerically. The application displays the function itself, secant and tangent line behavior, and numerical estimates of the derivative from the left and right sides of a selected point. The goal was to show how the derivative can be interpreted both visually and mathematically. The project combines calculus, numerical approximation, programming, and visualization. Instead of simply calculating a derivative, the application allows the user to see what happens as points move closer together and how the slope of a secant line approaches the slope of the tangent line.

A common problem when learning derivatives is understanding the connection between the derivative formula and what is happening on the graph of a function. The derivative is often introduced using the limit $f'(x) = \lim_{h \to 0} (f(x+h)-f(x))/h$. While this formula gives the mathematical definition of a derivative, it can be difficult to understand why it represents the slope of the tangent line. The purpose of this project was to create an application that demonstrates this relationship visually. The application needed to show the graph of a function, calculate approximate slopes, and help determine whether the function is differentiable at a selected point.

The derivative of a function at a point measures the instantaneous rate of change of the function. For a nonzero value of h, the difference quotient $(f(x+h)-f(x))/h$ represents the slope of

a secant line passing through two points on the function. As h becomes smaller, the second point moves closer to the first point. If the slope approaches a single value, that value represents the slope of the tangent line and therefore the derivative. The application uses this idea to estimate derivatives numerically.

A right-hand derivative can be approximated using (f(x+h)-f(x))/h, where h is a small positive number. A left-hand derivative can be approximated using (f(x)-f(x-h))/h. For a function to be differentiable at a point, the left-hand and right-hand derivatives should approach the same value. If these values do not agree, then the function is not differentiable at that point. This can happen when a function has a sharp corner, cusp, vertical tangent, or discontinuity.

The project was developed using R and the Shiny framework. Shiny makes it possible to create interactive web applications directly from R code. The application was designed to combine mathematical calculations with graphical output. A user can view a function and examine its behavior around a selected point. The program then calculates numerical derivative estimates and displays relevant graphical information. The numerical calculations rely on the difference quotient. Instead of calculating the limit symbolically, the program uses a small value of h to estimate the derivative computationally.

The application also includes a graphical representation of secant and tangent lines. A secant line passes through two points on the curve. As those two points get closer together, the secant line begins to approximate the tangent line. This makes the limit definition of the derivative more visual. Instead of only seeing the equation, a user can see how the geometry of the graph changes as the approximation improves.

The application demonstrates the difference between functions that are differentiable and functions that are not differentiable. For a smooth function, the numerical estimates from the left

and right sides of a point should be very close to each other. This indicates that the function has a consistent tangent slope at that location. For example, consider f(x)=x^2. Its derivative is f'(x)=2x, so at x=2 the derivative is 4. If the application calculates left-hand and right-hand derivative estimates using a sufficiently small value of h, both values should approach 4. Graphically, the secant lines also begin to approach the same tangent line.

A different situation occurs with a function such as f(x)=|x|. At x=0, the graph contains a sharp corner. The left-hand derivative is -1 while the right-hand derivative is 1. Since these two values do not agree, the function is not differentiable at x=0. This example demonstrates why examining both sides of a point is important. A function can be continuous at a point but still fail to be differentiable there. The application makes this difference easier to see because the user can compare the graphical shape of the function with the numerical derivative estimates.

The application uses numerical approximations rather than exact symbolic calculations, so the accuracy of the result depends partly on the selected value of h. If h is too large, the two points used in the secant calculation may be too far apart to give a good approximation of the tangent slope. If h is extremely small, floating-point precision can also become a problem. Computers have limits on how precisely decimal values can be represented. Because of this, making h smaller does not always guarantee a better answer.

Another limitation is that a graph alone cannot serve as a formal proof of differentiability. A graph may strongly suggest that a derivative exists, but mathematical reasoning is still necessary. The application should therefore be treated as a visualization and exploration tool rather than a replacement for analytical calculus. There are also limits to what types of functions can be entered or interpreted by the program, and more complicated expressions may require additional input validation or symbolic mathematics tools.

This project demonstrates mathematical modeling because it takes a theoretical mathematical concept and converts it into a computational process. The theoretical model begins with the limit definition of the derivative. That model is then translated into a numerical approximation that a computer can calculate. The application also demonstrates the relationship between analytical and numerical mathematics. Analytical methods are usually preferable when an exact solution is easy to calculate, while numerical methods become useful when the problem is more complicated or when an exact symbolic solution is difficult to obtain.

Visualization was an important part of the project because differentiability has both an algebraic and geometric meaning. The algebraic definition involves the limit of the difference quotient, while the geometric interpretation involves the slope of the tangent line. Combining these two ideas makes the concept easier to understand. The interactive graph allows users to observe how changing the selected point affects the derivative. Instead of looking at a single static example, users can explore different situations and observe the results themselves.

Overall, this project used R Shiny to create an interactive visualization of differentiability. The application combines calculus, numerical approximation, programming, and graphical analysis to demonstrate how derivatives behave. It shows how the derivative can be understood through both the limit definition and the geometric idea of a tangent line. The project also demonstrates the strengths and limitations of numerical methods. Numerical approximations can provide useful results and visual insight, but they do not completely replace analytical reasoning. Most importantly, the project helped connect mathematical theory with practical programming by turning an abstract calculus concept into something interactive and easier to explore.